

Antioxidant compounds EGB-761 and BN-52021 attenuate brain edema formation and hemeoxygenase expression following hyperthermic brain injury in the rat

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Summary

Role of carbon monoxide (CO) in hyperthermic brain injury induced brain pathology was examined in a rat model using immunohistochemistry of the hemeoxygenase-2 (HO-2) enzyme. Exposure of rats to 4 h heat stress at 38 °C resulted in profound hyperthermia, breakdown of the blood-brain barrier (BBB), brain edema formation, cell damage and expression of HO-2 in several brain regions. Pretreatment with potent antioxidant compounds EGB-761 and BN-52021 markedly reduced the HO-2 expression, BBB breakdown, brain edema formation and cell damage without attenuating the hyperthermic response. This effect was most marked in animals treated with EGB-761. These observations suggest that upregulation of HO-2 representing generation of CO plays important roles in hyperthermic brain injury, and oxidative stress seems to be one of the most important signals in inducing HO-2 expression in hyperthermia, not reported earlier.

Keywords: Hyperthermia; hemeoxygenase-2; carbon monoxide; brain injury; EGB-761; BN-52021; immunohistochemistry; edema formation; blood-brain barrier; Evans blue; radioiodine.

Introduction

The pathophysiology of hyperthermic brain injury includes alteration in the blood-brain barrier (BBB) permeability, edema formation and cell injury in the central nervous system (CNS) [17]. Several neurochemical mediators such as serotonin, prostaglandin, histamine and nitric oxide (NO) appear to participate in the pathological changes following hyperthermia induced by heat stress [22].

Since no single chemical compound is solely responsible for all the pathological changes seen in hyperthermic brain injury, it appears that other potential neurochemicals and endogenous factors are involved in brain pathology of hyperthermia. In this investigation, we examined the involvement of recently described signal transduction molecule, carbon monox-

ide (CO) in the pathophysiology of hyperthermic brain injury.

CO is a free radical gas which influences neuronal communication by modulating signal transduction in the CNS [8, 21]. The CO is a by product of the reaction when heme is converted into the antioxidant biliverdin and iron [7]. The enzyme responsible for conversion of heme into the CO is known as hemeoxygenase (HO) [7, 15] that occurs in two isoforms [7, 21]. The constitutive isoform of HO (HO-2) is present in the few normal nerve cells in the CNS [8, 17]. The inducible isoform of HO (HO-1) can be seen following several noxious stimuli to the brain such as ischemia, hypoxia and trauma [for review, see 15]. Recently, upregulation of HO in the CNS is described following a variety of chemical, hormonal and biological stressors [7, 15, 19]. The functional significance of HO upregulation is not clearly understood.

There are reasons to believe that CO is involved in several biological processes in the CNS [15, 17]. Previous reports from our laboratory suggest that hyperthermia caused by heat exposure is able to induce an upregulation of HO-2 enzyme in the CNS [13, 15]. This observation indicates that hyperthermia has the capacity to induce an upregulation of HO-2. However, whether increased production of CO as evidences with HO-2 expression in hyperthermia is associated with cell injury or cell survival is still not clearly known.

Hyperthermia is capable to induce release of free radicals probably due to an increase in the oxidative stress [4, 5] that contributes to hyperthermia induced cell damage in the CNS [5, 8]. Since CO is a free radical gas, it seems likely that its release in hyperthermia will induce cell damage. Thus, antioxidant compounds ca-

pable to inhibit or scavenge the free radicals will induce neuroprotection in hyperthermia [18, 19] and thus influence cellular stress mechanisms [17].

The present investigation was thus undertaken to investigate the influence of two antioxidant compounds EGB-761 and Gingkolide B (BN 52021) [1, 2, 9, 10] on hyperthermia induced HO expression, brain edema formation, cell injury and cellular stress. We have focused our attention on the expression of HO-2 and cell damage in a few selected brain regions in the control and heat stressed rats with or without treatment with the EGB-761 or BN-52021.

Materials and methods

Animals

Experiments were carried out on 56 male Sprague Dawley rats (body weight 110–140 g, age 9–10 weeks) housed at controlled room temperature at $21 \pm 1^\circ\text{C}$ with 12 h light and 12 h dark schedule. The rat food and tap water were supplied *ad libitum* before the experiments.

Heat exposure

Animals were exposed to heat in a biological oxygen demand (BOD) incubator maintained at 38°C (relative humidity 45–50%; wind velocity 20–26 cm/sec) for 4 h [11, 12, 14, 17]. This experimental condition is approved by the Ethical Committee of Uppsala University, Uppsala, Sweden. Rats kept at normal room temperature for 4 h without food and water were served as controls [11, 12].

Drug treatments

Two potent antioxidant substances EGB-761, which is scavenger of the free radicals [2, 9, 18] and the Gingkolide B [9, 19] were administered as described earlier. EGB-761 (50 mg/kg/day, p.o.) is a standardised extract from *Ginkgo biloba* leaves and Gingkolide B (BN-52021, 2 mg/kg/day, p.o.) (IPSEN-Pharma, France) were administered daily for 5 days. One dose of these compounds (0.3 ml/rat in distilled water) was also administered 30 min before the onset of heat exposure [18, 19].

Stress symptoms

Changes in the core body temperature, occurrence of salivation and the behavioural prostration were recorded using standard protocol as described earlier [11] (Table 1).

HO-2 immunohistochemistry

The HO-2 immunostaining was examined on free floating Vibratome sections (40 μm thick) obtained from different brain regions using a commercial monoclonal HO-2 antiserum (Stress Gene Ltd., Canada) as described earlier [13–15, 17]. The HO-2 antiserum was diluted 1:5000 and applied at room temperature under continuous agitation for 48 h. The immune reaction was developed by using peroxidase-antiperoxidase technique. In few sections, the primary antibody was omitted and the reaction product was developed [15]. The control, heat stressed, drug-treated controls and drug-treated heat stressed rats were processed in parallel [15, 16].

Table 1. Effect of EGB-761 and BN-52021 on stress symptoms and physiological variables in the control and 4 h heat stressed rats at 38°C

Type of Expt.	n	$^\circ\text{C}$ Rect. T	Saliva- tion	Prostra- tion	MABP	torr	Arterial pH	PaO ₂ torr	PaCO ₂ torr	BBB permeability		Brain edema formation	
										EBA mg %	^{131}I iodine %	Brain water %	% f
Control	5	–	nil	nil	110 $\pm$ 6		7.37 $\pm$ 0.03	81.34 $\pm$ 0.21	34.56 $\pm$ 0.33	0.24 $\pm$ 0.04	0.28 $\pm$ 0.03	74.34 $\pm$ 0.32	–
EGB-761 + control	5	–0.29 $\pm$ 0.05	nil	nil	108 $\pm$ 5		7.36 $\pm$ 0.08	80.98 $\pm$ 0.32	33.89 $\pm$ 0.76	0.18 $\pm$ 0.06	0.23 $\pm$ 0.05	74.18 $\pm$ 0.22	–
BN-52021 + control	5	+0.18 $\pm$ 0.08	nil	nil	112 $\pm$ 8		7.36 $\pm$ 0.04	81.02 $\pm$ 0.43	34.02 $\pm$ 0.44	0.26 $\pm$ 0.05	0.32 $\pm$ 0.08	74.47 $\pm$ 0.12	–
4 h HS	6	+3.48 $\pm$ 0.23***	++++	++++	78 $\pm$ 6***		7.34 $\pm$ 0.07	79.43 $\pm$ 0.98	32.89 $\pm$ 0.88	1.89 $\pm$ 0.24***	2.32 $\pm$ 0.28***	78.67 $\pm$ 0.34***	+18
EGB-761 + HS	5	+2.67 $\pm$ 0.18***a	++++	++++	82 $\pm$ 8***		7.33 $\pm$ 0.06	80.03 $\pm$ 0.76	33.04 $\pm$ 0.65	0.67 $\pm$ 0.11***a	0.89 $\pm$ 0.23***a	76.23 $\pm$ 0.12***a	+8
BN-52021 + HS	5	+2.89 $\pm$ 0.19***a	++++	++++	79 $\pm$ 6***		7.33 $\pm$ 0.09	79.89 $\pm$ 1.02	33.13 $\pm$ 0.87	0.86 $\pm$ 0.12***a	1.04 $\pm$ 0.18***a	76.98 $\pm$ 0.23***a	+12

Values are mean $\pm$ SD; MABP mean arterial blood pressure; HS heat stress; nil absent, +, ++, +++ severe, ++++ mild; *** $P < 0.001$, ** $P < 0.01$ compared to control; a $P < 0.05$, compared to h HS, Student's unpaired t-test.

Blood-brain barrier permeability and edema formation

The BBB permeability to Evans blue and ^{131}I -iodine was examined using standard procedures described earlier [12]. After counting the radioactivity, tissue pieces were kept in an incubator at 90°C for 72 h to determine their dry weight. The brain water content was calculated from the differences in the wet and dry weight of the samples as described earlier [13].

Cell injury

Tissue pieces from selected brain regions were processed for light and electron microscopy to study morphological changes using standard procedures [16].

Statistical analysis

Semiquantitative data were analysed using unpaired Student's t-test. A p-value less than 0.05 was considered as significant.

Results

Effect of EGB-751 and BN-52021 on stress response

Subjection of rats to 4 h heat stress resulted in profound hyperthermia (Table 1). Pretreatment with EGB-761 or BN-52021 slightly but significantly attenuated this hyperthermic response. However, the behavioural salivation or prostration following heat stress in these drug-treated rats were not affected (Table 1).

Effects of EGB-761 and BN-52021 on HO-2 expression

In normal animals only few cells showed mild HO-2 immunostaining distributed sporadically in the cortex, hippocampus, thalamus, hypothalamus in the brain stem (Fig. 1). Subjection of rats to 4 h heat stress resulted in massive upregulation of HO-2 expression in many brain regions. This was most extensive in the thalamus and hypothalamus. The nerve cell cytoplasm showed most immunostaining however, occasional nerve cell nuclei were also stained (Fig. 2). Pretreatment with EGB-761 or BN-52021 significantly reduced heat stress induced upregulation of HO-2 expression in several brain regions (Fig. 2). The influence of EGB-761 in attenuating heat stress induced HO-2 expression was most pronounced compared to the BN-52021 (Figs. 1 and 2).

BBB permeability and edema formation

Subjection of rats to 4 h heat stress resulted in marked extravasation of Evans blue albumin and ra-

dioactive iodine in the CNS (Table 1). This leakage of protein tracers was significantly attenuated by pretreatment with EGB-761 or BN-52021 (Ginkgolide B) (Table 1). However, the effect of EGB-761 is far more superior in reducing the hyperthermia induced protein leakage in the brain compared to the Ginkgolide B (BN 52021) alone (Table 1). These compounds are also able to thwart edema formation following heat stress (Table 1). This effect is most pronounced in the EGB-761 treated rats compared to the BN 52021 treatment (Table 1).

Cell injury in heat stress

Hyperthermia resulted in nerve cell distortion, edema, sponginess and general expansion of the brain (Fig. 2). These effects are significantly attenuated by pretreatment with EGB-761 or BN-52021 (Fig. 2). However, the effects of EGB-761 are far more superior than that of BN-52021 in reducing cell injury in the animals subjected to 4 h heat stress (Fig. 2).

Discussion

The present findings clearly show that heat stress induced hyperthermic brain injury has the capacity to induce an upregulation of HO-2 in several brain regions. This observation is in line with the idea that CO plays important roles in neuronal injury [13, 15]. There are several lines of evidences which suggest that up-regulation of HO-2 is associated with increased production of CO [7, 21]. Thus, it seems likely that an abnormal production of CO during hyperthermia is responsible for cell injury. A marked reduction in HO-2 expression and cell injury in heat stress by prior treatment with EGB-761 and BN-52021 in present study further confirms this idea.

EGB-761 is a well known scavenger of free radicals [9, 18, 19]. On the other hand, Ginkgolide B (BN 52021) one of the active constituent of the EGB-761 is a powerful antioxidant [9, 10]. Thus, EGB-761 and BN-52021 are able to attenuate heat stress induced generation of free radicals either by scavenging them or by attenuating the oxidative and/or cellular stress mechanisms [18–20]. This could be one of the most important mechanisms responsible for reduction in cell changes caused by these compounds in hyperthermia. Interestingly, these compounds are also able to attenuate HO-2 expression in heat stress. This observation suggests that the oxidative stress and associated free

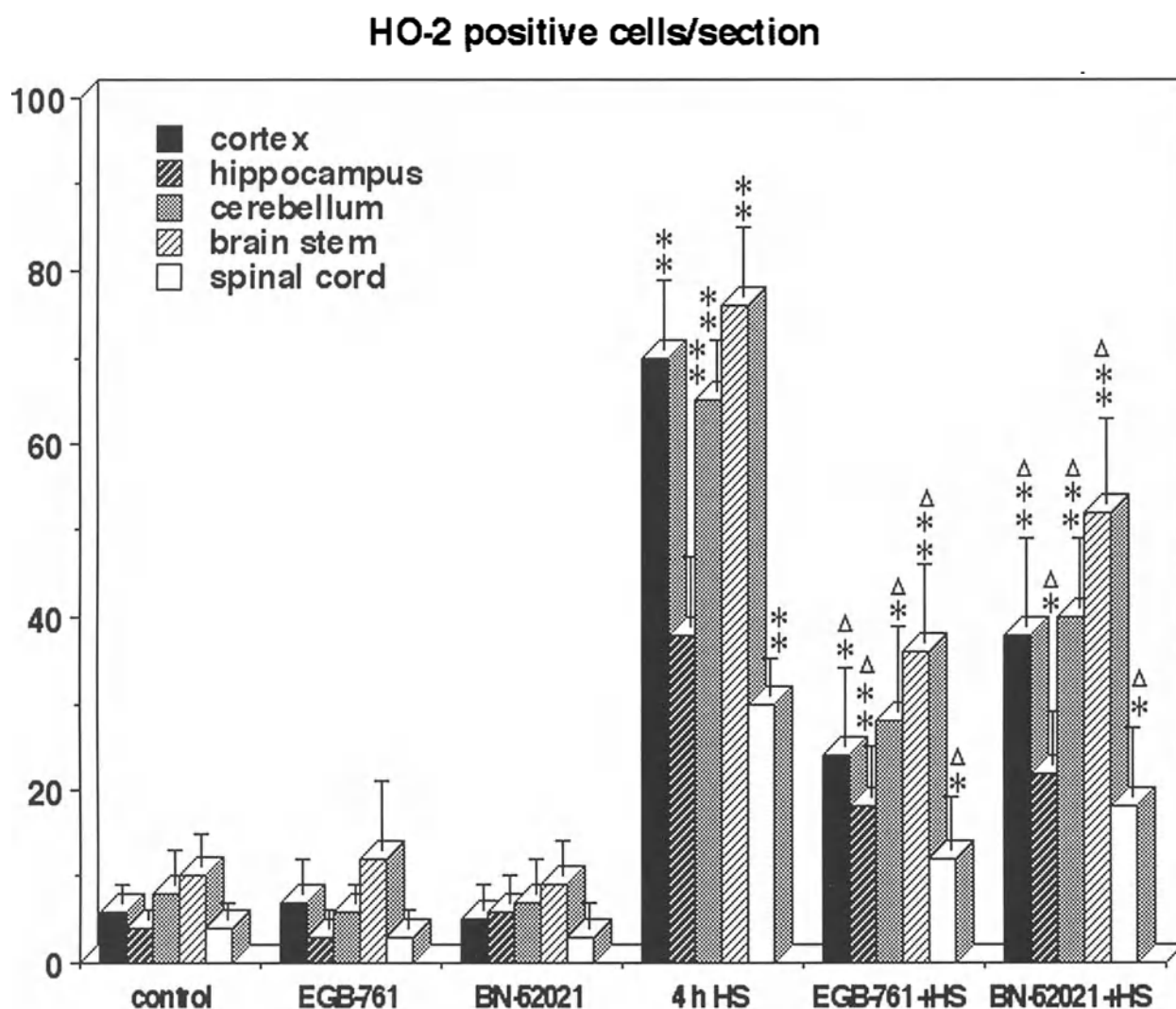


Fig. 1. Semiquantitative analysis of HO-2 immunostaining in control and heat stressed animals and their modification with EGB-761 or BN-52021. * = $P < 0.01$; ** = $P < 0.001$ compared with the control, D = $P < 0.01$ compared with 4 h heat stress (HS), Student's unpaired t-test

radical formation in hyperthermia is important in inducing HO-2 expression in the CNS.

Oxidative stress induces upregulation of nitric oxide synthase following trauma, ischemia or hypoxia [21]. Previous reports from our laboratory suggest that pretreatment with EGB-761 and BN-52021 reduce the upregulation of both constitutive and inducible isoforms of nitric oxide synthase [19]. This suggests that oxidative stress and free radical formation induce NOS upregulation in hyperthermia [18, 19, 21]. However, oxidative stress induced upregulation of HO is not well documented in the literature. Our results, probably for the first time using EGB-761 and BN-52021 in hyperthermia demonstrate that oxidative stress is somehow responsible for increased CO production as evident from an upregulation of HO-2 immunostaining.

Taken together, these observations suggest that in hyperthermia NOS and HO seem to work in synergy [6] to induce cell damage.

Upregulation of CO as seen by increased immunostaining of HO-2 induces cell damage in spinal cord trauma [20]. Prevention of HO-2 upregulation in trauma is associated with neuroprotection. Since basic mechanisms of cell injury caused by trauma, ischemia or hypoxia appears to be quite similar in nature, it is likely that less intense immunoreaction of HO-2 in the EGB-761 or BN-52021 treated rats is related with mild cell damage in hyperthermia.

Hyperthermia is associated with severe cellular stress [4, 17] as seen by marked upregulation of heat shock proteins (HSP 72 kD) in the CNS [16, 17]. Cellular stress is caused by oxidative stress associated with

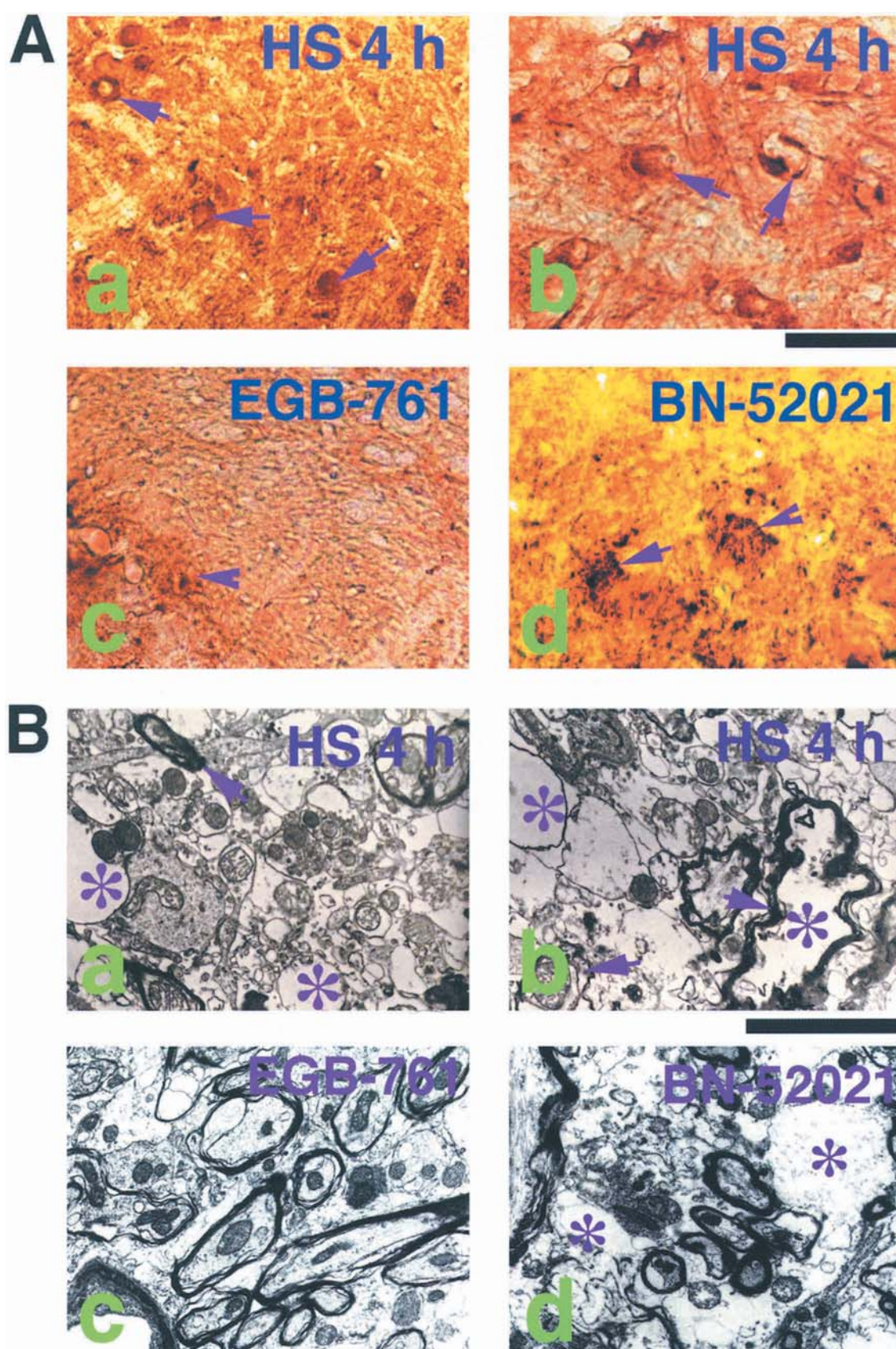


Fig. 2. (A) A representative example of HO-2 immunostaining in the spinal cord on one heat stressed rat (a, b) and its modification with EGB-761 (c) and BN-52021 (d) pretreatment. Bar = 50 μ m. (B) Low power electron micrograph show pronounced cell injury in heat stressed rat (a, b) and its modification with EGB-761 (c) and BN 52021 (d) treatments. Marked neuroprotective effects are apparent in the EGB-761 treated stressed rat compared to the BN-52021 (bar = 1 μ m) (data modified after Westman *et al.*, 2000)

free radical formation [3, 5]. Since EGB-761 and BN-52021 are able to reduce HSP 72 kD expression in hyperthermia [22], it appears that these compounds are interfering with the molecular mechanisms of cellular stress phenomena. Obviously, a less stress reaction in heat-stressed animals will result in mild release of free radicals.

Heat stress induced hyperthermia alone is not responsible for HO-2 expression or cell injury. Thus, EGB-761 or BN-52021 pretreatment resulted in a mild reduction in hyperthermia and no considerable reduction in the stress symptoms was observed in these drug-treated stressed group. This suggests that the magnitude of hyperthermia is not directly related with the severity of the heat stress induced behavioural symptoms. However, further studies are needed to clarify this point.

In conclusion, our study demonstrates that the antioxidants EGB-761 and its constituent Ginkgolide B (BN 52021) has the capacity to attenuate heat stress induced HO-2 upregulation. This indicates that these compounds exert their neuroprotective effects in hyperthermic brain injury by influencing CO production, not reported earlier. However, further studies are needed to understand the mode of action of these compounds in attenuating HO-2 expression in hyperthermia.

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References

1. Bastianetto S, Zheng WH, Quirion R (2000) The Ginkgo biloba extract (EGB 761) protects and rescues hippocampal cells against nitric oxide-induced toxicity: involvement of its flavonoid constituents and protein kinase C. *J Neurochem* 74: 2268–2277
2. Drieu K, Vrancks R, Benassayad C, Haourigi M, Hassid J, Yoa RG, Rapin JR, Nunez EA (2000) Effect of the extract of ginkgo biloba (EGB 761) on the circulating and cellular profiles of polyunsaturated fatty acids: correlation with the anti-oxidant properties of the extract. *Prostaglandins Leukot Essential Fatty Acids* 63: 293–300
3. Flanagan SW, Mooseley PL, Buettner GR (1998) Increased flux of free radicals in cells subjected to hyperthermia: detection by electron paramagnetic resonance spin trapping. *FEBS Lett* 431: 285–286
4. Freeman ML, Spitz DR, Meredith MJ (1990) Does heat shock enhance oxidative stress? Studies with ferrous and ferric iron. *Radiat Res* 124: 288–293
5. Hedlund S, Sharma HS, Sjöquist P-O, Westman J (1999) A new antioxidant compound H-290/51 attenuates heat shock protein (HSP 72 kD) response, edema and cell injury following acute heat exposure. An experimental study using light and electron microscopy in the rat. *J Therm Biol* 24: 409–414
6. Kim CK, Rivier CL (2000) Nitric oxide and carbon monoxide have a stimulatory role in the hypothalamic-pituitary-adrenal response to physico-emotional stressors in rats. *Endocrinology* 141: 2244–2253
7. Maines MD (1997) The heme oxygenase system: a regulator of second messenger gases. *Annu Rev Pharmacol Toxicol* 37: 517–554
8. Miura T, Muraoka S, Fujimoto Y (1998) Temperature-dependent lipid peroxidation of rat brain homogenate. *Res Commun Mol Pathol Pharmacol* 100: 117–128
9. Pietri S, Seguin JR, d'Arbigny P, Drieu K, Culcasi M (1997) Ginkgo biloba extract (EGB 761) pretreatment limits free radical-induced oxidative stress in patients undergoing coronary bypass surgery. *Cardiovasc Drugs Ther* 11: 121–131
10. Sastre J, Millan A, Garcia de la Asuncion J, Pla R, Juan G, Pallardo, O'Conner E, Martin JA, Droy-Lefaix MT, Vina J (1998) A Ginkgo biloba extracts (EGB 761) prevents mitochondrial aging by protecting against oxidative stress. *Free Radic Biol Med* 24: 298–304
11. Sharma HS, Dey PK (1986) Probable involvement of 5-hydroxytryptamine in increased permeability of blood-brain barrier under heat stress. *Neuropharmacology* 25: 161–167
12. Sharma HS, Dey PK (1987) Influence of long-term acute heat exposure on regional blood-brain barrier permeability, cerebral blood flow and 5-HT level in conscious normotensive young rats. *Brain Res* 424: 153–162
13. Sharma HS, Alm P, Westman J (1997a) Upregulation of hemeoxygenase-II in the rat spinal cord following heat stress. *Thermal physiology*. In: Nielsen-Johannsen B, Nielsen R (eds) *The August Krogh Institute, Copenhagen, 1997*, pp 135–138
14. Sharma HS, Westman J, Cervós-Navarro J, Nyberg F (1997b) Role of neurochemicals in brain edema and cell changes following hyperthermic brain injury in the rat. *Acta Neurochir (Wien) [Suppl]* 70: 269–274
15. Sharma HS, Alm P, Westman J (1998a) Nitric oxide and carbon monoxide in the pathophysiology of brain functions in heat stress. *Prog Brain Res* 115: 297–333
16. Sharma HS, Westman J, Nyberg F (1998b) Pathophysiology of brain edema and cell changes following hyperthermic brain injury. *Prog Brain Res* 115: 351–412
17. Sharma HS (1999) Pathophysiology of blood-brain barrier, brain edema and cell injury following hyperthermia: New role of heat shock protein, nitric oxide and carbon monoxide. An experimental study in the rat using light and electron microscopy. *Acta Universitatis Upsaliensis* 830: 1–94
18. Sharma HS, Drieu K, Alm P, Westman J (1999) Upregulation of neuronal nitric oxide synthase, edema and cell injury following heat stress are reduced by pretreatment with EGB-761 in the rat. *J Therm Biol* 24: 439–446
19. Sharma HS, Drieu K, Alm P, Westman J (2000a) Role of nitric oxide in the blood-brain barrier permeability, brain edema and cell damage following hyperthermic brain injury. An experimental study using EGB-761 and Ginkgolide B pretreatment in the rat. *Acta Neurochir (Wien) [Suppl]* 76: 81–86
20. Sharma HS, Alm P, Sjöquist P-O, Westman J (2000b) A new antioxidant compound H-290/51 attenuates upregulation of constitutive isoform of heme oxygenase (HO-2) following trauma to the rat spinal cord. *Acta Neurochir (Wien) [Suppl]* 76: 153–157

21. Snyder SH, Ferris CD (2000) Novel neurotransmitters and their neuropsychiatric relevance. *Am J Psychiatry* 157: 1738–1751
22. Westman J, Drieu K, Sharma HS (2000) Antioxidant compounds EGB-761 and BN-52021 attenuate heat shock protein (HSP 72 kD) response, edema and cell changes following hy-

perthermic brain injury. An experimental study using immunohistochemistry in the rat. *AminoAcids* 19: 339–350

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